

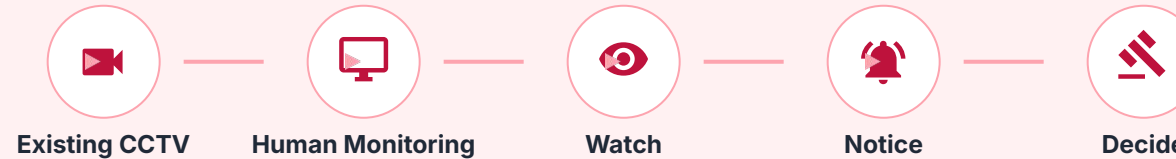
TITLE PAGE

- **Problem Statement ID** – 26187
- **Problem Statement Title-** AI-Based Intelligent Video Analytics Platform for Border Surveillance using existing CCTV Infrastructure.
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal)-** DFS404

THE CURRENT LANDSCAPE

Manual. Reactive. Inefficient.

The raw CCTV video remains unmonitored or poorly analyzed in real time.



THE CORE PAIN POINTS



Continuous Monitoring

Many feeds compete for attention.



Low-Light Uncertainty

Night scenes are harder for generic detectors..



Specialized Analytics

Advanced analytics require brand new, expensive hardware. Software scale-out is blocked.

THE INSIGHT & OUR SOLUTION

Inserting intelligence directly between raw data streams and human operators.

Existing CCTV

Legacy IP or analog cameras remain untouched.



IBVAP

Intelligent Video Analytics

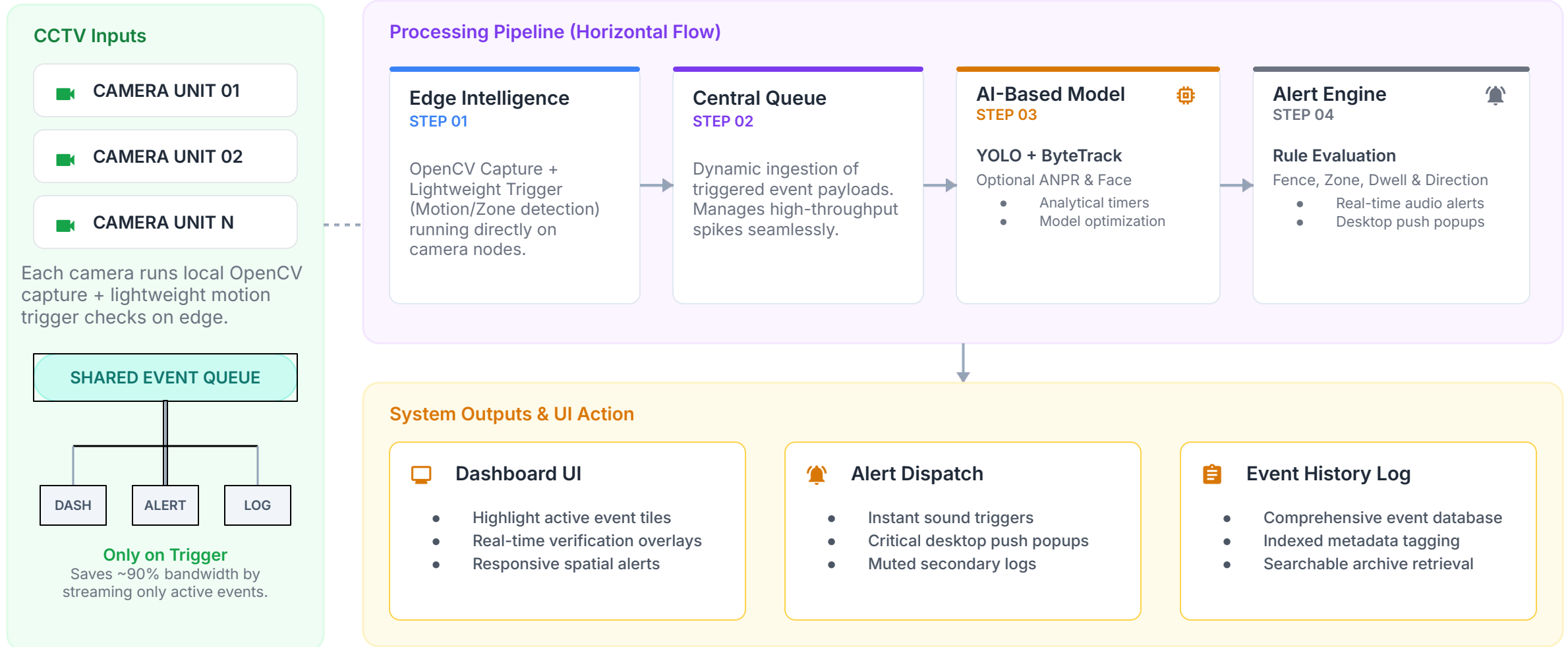
Pure software layer parsing motion and targets on edge.



Actionable Events

Only what matters. When it matters.

"We are not replacing the operator. We are reducing the video they need to watch."





DEPLOYMENT ROADMAP

1. Pilot (1-5 Cams / 1 BOP)

Prove detection & alert accuracy against real terrain/lighting. Modest entry risk for SIH buyer.

2. Full Site Coverage

Scale to all cameras at post. Event-driven queue optimizes compute resource allocation dynamically without linear scaling.

3. Multi-BOP Replication

Duplicate proven independent edge-boxes across new strategic locations without system redesign or complex integration.



BUSINESS & PROCUREMENT

1. One-Time Setup

On-site calibration (fences, restricted zones, dwell thresholds tailored to geography) and core C2 integration services.

2. Annual License (Recurring)

Funds continuous deep-learning model improvements for low-light & challenging edge cases as operational footage builds.

3. Annual Support Plan

Operator training programs and ongoing spatial rule recalibration as BOP terrain, barriers, or local hardware configurations change.



COST ADVANTAGE

1. Existing Infrastructure Retrofit

Skip expensive new camera sensor deployment or hardware swaps. Retain legacy IP and analog feeds seamlessly.

2. Hardware Minimization

One smart processing box per site. Feeds run quiet and conserve bandwidth until motion triggers the heavier AI models.

Saves ~90% compared to hardware replacement

SUSTAINABILITY & SYSTEM CALIBRATION (WHY THIS REMAINS VIABLE)

Deploy & Adapt: Proves capability in brutal field conditions first. Low incremental costs drive easy expansion to adjacent cameras.

Continuous Accuracy: Annual software renewals fund retraining of detection weights against seasonal changes, foliage growth, and new environmental noise.

Operational Support: Keeps site bounds fully calibrated to prevent system accuracy degradation over time as physical guard posts evolve.

"Don't replace the cameras. Make them useful."

Proposed Deployment Timeline: 3-6 Month Pilot Phase before wide-scale BOP licensing decision.

MANUAL VS IBVAP

Today: One operator scans 8-16 active feeds over an 8-hour shift. Attention degrades rapidly; critical security events are missed.

With IBVAP: IBVAP filters surveillance noise, helping operators focus on critical events.

WHY NOT SIMPLER TECH?

Beyond Motion Detection: Basic motion sensors act like smoke alarms triggered by steam. IBVAP understands spatial and context rules.

The Video Spam Filter: It operates as a highly specialized filter, removing up to 95% of irrelevant movement that doesn't require a human eye.

NO NEW HARDWARE

Zero Camera Touch: Existing IP cameras stay exactly where they are. We do not require proprietary physical upgrades.

Legacy Friendly: Analog feeds integrate seamlessly through existing DVRs/encoders, adapting to current infrastructure.

POSITIONING AGAINST NATIONAL SYSTEMS (PRE-EMPTING OBJECTIONS)

The National Precedent: High-profile defense systems prove this direction works. The Army runs ~140 AI surveillance networks along the LoC/LAC, and the MHA utilizes BOLD-QIT/CIBMS along unfenceable riverine borders. These are expensive, multi-sensor deployments built for critical frontiers.

Our Narrower, High-Impact Focus: *"You're right — those prove the tech works. Our contribution is targeted."* IBVAP is built for the much larger tier of ordinary Border Outposts (BOPs) that already have CCTV cameras installed but possess zero software intelligence. It is a rapid, pure-software retrofit.

EDGE-CASE STRIP & SAFETY HANDLING

Very Dark
lower confidence



Occlusion
tracking may break



Blurred Plate
OCR may fail



Bad Camera
info unavailable

Core Design Philosophy: Low confidence triggers human verification, never false certainty.

- [Ultralytics\(YOLO\) - Github](#)
- [YOLO Docs](#)
- [LLVIP: A Visible-infrared Paired Dataset for Low-light Vision](#)
- [ByteTrack: Multi-Object Tracking by Associating Every Detection Box](#)
- [CLAHE-Based Low-Light Image Enhancement for Robust Object Detection in Overhead Power Transmission System](#)
- [Real-Time Automatic License Plate Recognition](#)